Harshavarthan Venkatesan

EMBEDDED ANALYST

Email: venkyharsha2005@gmail.com
+916383006873
Website: www.reallygreatsite.com

PROFESSIONAL SUMMARY

Enthusiastic Artificial Intelligence and Data Science student with hands-on experience in embedded systems, AI applications, and Microcontroller based projects. Proficient in Python, C, and web technologies, with strong problem-solving and design thinking skills. Passionate about integrating hardware and software to develop intelligent, real-world IoT and AI solutions.
A fast learner and dedicated team player with a drive to explore innovation through technology.

EDUCATION

Lisieux Mat Hr Sec School
Combatore | Computer Science. *2009 — 2023*

• Scored 100% in 10th and 91% in 12th
2009 — 2023

- School English topper in 12th
- Active in co-curricular activities; received Governor's Award in Scouts Black belt holder in Karate
- Represented zone level in Badminton

Kumaraguru College of Technology. 2023 — 2027
Combatore Btech Artificial intelligence and Data science

- Cgpa- 7.2 (5th sem)
- Gained hands-on experience in cloud computing, embedded systems, AI model training, Business Intelligence and IoT solutions.
- Developed knowledge in neural networks, EDA and reinforcement learning
- Represented India as part of the only Indian team at the Monaco
- Energy Boat Challenge

WORK EXPERIENCE

Embedded and Data Visualization engineer Sep 2024 — Sep 2025
Team Sea Sakthi - Coimbatore

- Contributed to embedded systems development including, sensor integration, circuit design, designing and deployment
- Expertised in embedded software and tools Built data visualization solutions across web platforms and AR smart glasses.
- Developed an FPV based on the pilot view to the ground station and autonomous river cleaning robot as part of Team Sea Sakthi

Innovation Engineer Trainee. **Jan 2026 - Jun 2026**
Forge Innovation & Ventures - Coimbatore

- Trained as an Innovation Engineer Trainee at Forge Innovation & Ventures, focusing on real-world problem solving and product development
- Gained hands-on experience in innovation processes, startup fundamentals, and building scalable solutions
- Selected as the scholarship recipient for the protosem in Forge Innovation & Ventures.

SKILLS

- Embedded Systems
- Data Visualization and Analysis
- Cloud architecture
- Model training
- Business Intelligence

PROGRAMMING LANGUAGES

- HTML, CSS, JS
- Python
- C, Embedded C
- Java

LANGUAGES

- Tamil
- English
- Geraman (A2*)
- Hindi(Praveen)
- Malayalam
- Kanada(Limited working proficiency)

PROJECTS

YALI 4.0 – Team Sea Sakthi | Monaco Energy Boat Challenge

- Managed embedded systems operations including telemetry , data acquisition and visualization
- • Integrated sensors, and planned electronic circuits
- Worked extensively with microcontrollers for real-time system control

AUTONOMOUS RIVER CLEANING BOT | Team Sea Sakthi

- Embedded system development and overall operations of the project
- Implemented autonomy using ArduPilot with Pixhawk
- Designed system architecture, selected components, and developed circuit plans

FPV & AR VISUALIZATION SYSTEM | Team Sea Sakthi

- Developed FPV system where camera direction follows pilot view, transmitting real-time video to ground station using ESP32
- Designed and deployed system architecture for real-time communication
- Part of Building AR-based visualization (Unity + XREAL glasses) displaying battery, voltage, thrust, speed, and drainage metrics
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TEAM SEA SAKTHI WEBSITE

- Developed and deployed official team website using HTML, CSS, and JavaScript
- Integrated advanced animations using tools like GSAP, BOLT
- Independently handled design, development, and deployment
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AGRI MARKET ANALYSIS DASHBOARD

- Built AI-powered dashboard to analyze crop price trends and forecast future prices
- Implemented Neural Prophet model with performance metrics (RMSE, MAE, Loss)
- Developed full-stack solution (HTML, CSS, JS, Flask, SQLite) and deployed on Vercel

CERTIFICATIONS AND LICENSES

- Introduction to Cybersecurity Tools & Cyberattacks by IBM
- Cybersecurity Foundations by Malcolm Shore
- Data visualization by University of Illinois Urbana-Champaign
- Machine Leaning with Python: Foundations by Frederick Nwanganga
- Power BI Data Modelling Basics by Microsoft
- Introduction to Data Mining by skillup
- Power BI Business Scenarios with Hands on use case by Udemy
- Microstratergy for Business Intelligence - The Complete Course by Udemy
- Intra Nova CTF by iQube Kumaraguru

ACHIEVEMENTS

- Monaco Energy Boat Challenge 2025 - overall 8th out of 22
- Certificate of Recognition by Kumaraguru Institutions for academic and Cocurricular pursuits 2025
- Scholarship recipient for Protosem by Forge Innovation & Ventures

Honors:

- Invited by the Ambassador of India to France and Monaco for felicitating Team Sea Sakthi in the Indian Embassy Paris, France
- Attended the Conference in International Energy Agency Paris, France